AIRBUS

Airbus Material Specification

Precipitation Hardening Martensitic Stainless Steel
X5CrNiCu15-5 (15-5PH)
Solution Treated and Precipitation Hardened
1070 MPa ≤ Rm ≤ 1200 MPa
Forging
a ≤ 150 mm

Export Control:

FR Export Control - Not Listed. This Item is not listed against the EC regulations in the EU/FR.

US EC - No US content. This Item does not contain any U.S. origin ITAR or EAR content.

Contents

1 Scope

This Material Specification defines the requirements of Precipitation Hardening Martensitic Stainless Steel X5CrNiCu15-5 (15-5 PH) Solution Treated and Precipitation Hardened, Rm ≥ 1070 MPa, thickness a ≤ 150 mm for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

No typical application.

3 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS01-08-000	Technical Specification - Steel and Steel Alloys - Forged parts
AMS2315	Determination of delta ferrite content
ASTMA604	Standard Practice for Macroetch Testing of Consumable Electrode Remelted Steel Bars and Billets.
ASTME975	Standard practice for x-ray determination of retained austenite in steel with near random crystallographic orientation
ASTME1268	Standard practice for assessing the degree of banding or orientation of microstructures
EN4500-5	Aerospace series- Metallic materials- Rules for drafting and presentation of materials standards – Part 5: Specific

4 Requirements

4.1 General

Specification values are minimum except where stated.

4.2 Specific

The data requirements given in the tables are according to the relevant EN4500-5.

5 Designation

6 Technical Specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS01-08-000.

1	Material designation				X5CrNiCu15-5 (15-5 PH)

No	Chemical Composition %	Element	C	Si	Mn	P	S	Cr	Ni	Mo	Cu	Nb+ Ta	Fe
2	Chemical Composition %	min.	-	-	-	-	-	14,3	4,50	-	2,5	5.C	Base
		max.	0,07	1,00	1,00	0,025	0,005	15,3	5,50	0,50	4,50	0,45	

3	Method of melting				Degassing AOD or VOD + Consumable Electrode Melting
4	Form Method of production Limit dimension (mm)				Die or hand forging From forging stock in accordance with AIMS01-08-000 a ≤ 150mm
5	Technical specification				AIMS01-08-000
6	6.1 Delivery condition				Solution treated and precipitation hardened
	and heat treatment				1025°C ±10°C ≤ θ ≤ 1050°C ±10°C / Air Cooling or faster to θ ≤ 20°C Artificial ageing at 540°C≤θ≤570°C /4hrs 1) in a furnace at ±5°C / Air or water cooling
	6.2 Delivery condition code				According to drawing set
7	Use condition and heat treatment				Solution treated and precipitation hardened
8	Sample Test piece Heat treatment				Use condition
9	Dimension concerned		a	mm	a ≤ 150
10	Thickness of cladding on each face			%	-

					L	LT	ST
11	Direction of test piece				L	LT	ST
12	TENSILE	Temperature	θ	°C	Room temperature		
13		Proof stress	$R_{p0,2}$	MPa	≥ 1000		
14		Strength	Rm	MPa	1070 ≤ Rm ≤ 1200		
15		Elongation	A	%	≥ 12	≥ 12	≥ 8
16		Reduction of area	Z	%	≥ 60	≥ 60	≥ 50
17	Hardness		HB	-	321 ≤ HB ≤ 375		
18	Shear strength		Rc	MPa	-		
19	Bending		k	-	-		

20	Impact strength	k	-	Direction of test piece		L-T		T-L	
				Test temperature		+20°C	-30°C	+20°C	-30°C
				Energy KV	J	≥ 80	≥ 35	≥ 56	≥ 25

21	CREEP	Temperature	θ	°C	-
22		Time		h	-
23		Stress	σ_a	MPa	-
24		Elongation	a	%	-
25		Rupture stress	σ_R	MPa	-
26		Elongation at rupture	A	%	-
27	Notes (see Line 98)				1), 2), 3) and 4)

30	Microstructure	1	ASTME1268			
		3	Inspection plane according to drawing set			
		7	Homogeneous and fine martensitic structure			
	Ferrite δ content	1	AMS2315			
		7	Ferrite δ≤ 2%			
	Austenite γ content	1	ASTME975			
		3	X-Ray			
		7	Austenite γ ≤ 10%			
34	Grain size	-	See AIMS01-08-000			
		3	Inspection plane according to drawing set			
		7	ASTM G ≥ 6 some ASTM G ≥ 4 allowed			
40	Fracture toughness (K_{1C}) [2]	-	See AIMS01-08-000			
		3	All directions			
		6	-30°C	Room temperature		
		7	≥ 100 MPa√m	≥ 120 MPa√m		
44	External defects	-	See AIMS01-08-000			
46	Fatigue	-	See AIMS01-08-000			
		3	Notched hole T type specimen			
		7	The curve of the results shall not lie below the reference curve defined by the following data			
			σ_{max} in MPa	Number of cycles	R	Kt
			570	4,0E04	0,1	2,3
			450	1,0E05		
			370	1,0E06		
50	Cleanness inclusion content [3] (micro cleanness)	-	See AIMS01-08-000			
		7	Cat 4 Method D			
51	Macrostructure – Grain flow	-	Grain flow according to drawing set and AIMS01-08-000			
	Macrostructure – defects [3]	-	ASTMA604			
		7	Class	Condition	Severity	
			1	Freckles	A	
			2	White spots	A	
			3	Radial segregation	A	
			4	Ring pattern	B	
61	Internal defects	-	See AIMS01-08-000			
		7	Class AA			
68	Density (for information)	-	7,8 g/cm^3			

<table>
<tr><td rowspan="9">71</td><td rowspan="9">Crack propagation</td><td>-</td><td colspan="3">See AIMS01-08-000</td></tr>
<tr><td>3</td><td colspan="3">CTW75B12 specimens 4)</td></tr>
<tr><td rowspan="7">7</td><td colspan="3">The crack propagation rates should not be greater than the following data</td></tr>
<tr><td>ΔK in MPa√m</td><td>da/dN (mm/cycle)</td><td>R</td></tr>
<tr><td>20</td><td>1,00E-04</td><td rowspan="5">0,1</td></tr>
<tr><td>30</td><td>2,50E-04</td></tr>
<tr><td>40</td><td>4,00E-04</td></tr>
<tr><td>50</td><td>7,00E-04</td></tr>
<tr><td>60</td><td>1,00E-03</td></tr>
<tr><td>-</td><td>Ferromagnetic</td><td>7</td><td colspan="3">Yes</td></tr>
<tr><td>82</td><td>Batch uniformity</td><td></td><td colspan="3">AIMS01-08-000</td></tr>
<tr><td>98</td><td>Notes</td><td>-</td><td colspan="3">1) Tolerance on time of heat treatment stated in AIMS01-08-000

2) K_Q results are acceptable to satisfy minimum K_{1C} requirement only if the criteria's $P_{max}/P_Q \le 1{,}1$ and/or $W\text{-}a < 2{,}5(K_Q/R_{p0,2})^2$ are found invalid.

K_Q values invalid on the basis of criteria's other than those

listed above shall not be used to satisfy the minimum K_{1C}

requirement of line 40.

3) Macrostructure defects and cleanness inclusion content shall be checked at billet stage to ensure forging stock will yield products meeting with requirements.

4) Smaller specimens shall only be used where the dimensions of the forging preclude the use of CTW75 specimens. To be agreed between manufacturer and purchaser.</td></tr>
<tr><td>99</td><td>Typical use</td><td>-</td><td colspan="3">-</td></tr>
<tr><td>100</td><td>Qualification</td><td></td><td colspan="3">Frequency of qualification testing is included in AIMS01-08-000</td></tr>
</table>

Record of Revisions

Issue	Clause modified	Description of Modification
1 11.2012		New Standard
2 03.2024	All Chap 6	New template Modification of K1C values at room temperature